

DIGITAL IMAGE PROCESSING



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DIGITAL IMAGE AQUISITION

Is the foundational process of capturing physical, analog, or visual information from the real world (such as light, X-rays, or heat) and converting it into a discrete, numerical 2D array of pixels, or a digital image, for computer processing.

DIGITAL IMAGE ACQUISITION DEVICES

The imaging acquisition devices are composed of imaging modalities and acquisition gateway computers.

DIGITAL IMAGE ACQUISITION Contd

Examples

- **Cameras** A camera is an instrument used to capture and store images and videos, either digitally via an electronic image sensor, or chemically via a light-sensitive material such as photographic film
- **Fingerprint Scanners are** biometric security devices that capture, record, and analyze the unique ridge and valley patterns on a person's finger for identification or authentication
- **Iris Scanners are** advanced biometric devices that identify individuals by capturing and analyzing the unique, intricate patterns in the colored ring of the eye

DIGITAL IMAGE ACQUISITION Contd

Examples

- **Magnetic resonance imaging machines** are non-invasive, tube-shaped scanners that use powerful magnets, radio waves, and computers to produce highly detailed, 3D images of soft tissues, organs, and bones without using radiation
- **Radiography Machine** X-Ray is a medical imaging device that uses controlled, high-energy electromagnetic radiation to produce black-and-white images of the body's internal structures, primarily bones and dense tissues

DIGITAL IMAGE ACQUISITION Contd

Examples

- **Computed tomography machines** are advanced diagnostic imaging devices that use rotating X-ray equipment and computer algorithms to generate detailed, cross-sectional, 3D images (“slices”) of bones, blood vessels, and soft tissues. Faster and more detailed than X-Rays, for cancer
- **Echocardiogram Machines** are non-invasive, medical imaging devices that use high-frequency sound waves (ultrasound) to produce real-time, moving images of the heart's structure, chambers, valves, and blood flow

THE PINHOLE CAMERA MODEL

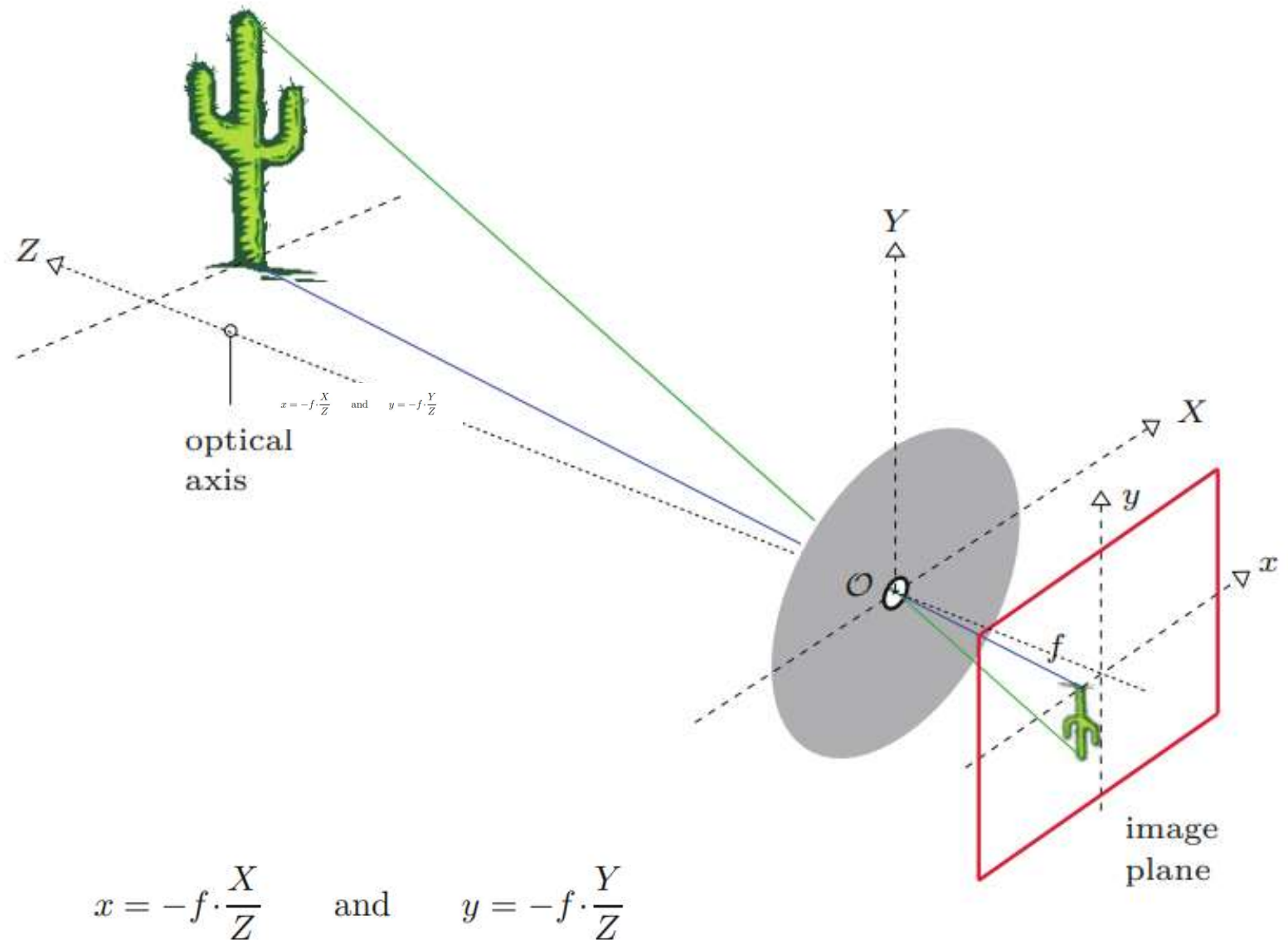
- The pinhole camera is one of the simplest camera models
- It has been in use since the 13th century, and is known as the “Camera Obscura”.
- The pinhole cameras have no practical use today except for hobbyists
- They are a useful model for understanding the essential optical components of a simple camera

Concept

- It consists of a closed box with a small opening on the front side through which light enters, forming an image on the opposing wall
- The light forms a smaller, inverted image of the scene

THE PINHOLE CAMERA MODEL

Geometry of the pinhole camera. The pinhole opening serves as the origin ($\mathcal{O}$) of the 3D coordinate system (X, Y, Z) for the objects in the scene. The optical axis, which runs through the opening, is the Z axis of this coordinate system. A separate 2D coordinate system (x, y) describes the projection points on the image plane. The distance f ("focal length") between the opening and the image plane determines the scale of the projection.



THE PINHOLE CAMERA MODEL Cont'd

- For a fixed image, a small f (i.e., short focal length) results in a small image and a large viewing angle, just as occurs when a wide-angle lens is used.
- while increasing the “focal length” f results in a larger image and a smaller viewing angle.

THE PINHOLE CAMERA MODEL Cont'd

2 Important properties of this theoretical model are

- that straight lines in 3D space always appear straight in 2D projections and
- that circles appear as ellipses

IMAGE CONVERSION TO DIGITAL

Characteristics of projected image

- Two-dimensional (2D),
- Time-dependent,
- Continuous distribution of light energy.

Conversion of this image into a digital image on the computer, requires three main steps

IMAGE CONVERSION STEPS

- Step 1: Spatial sampling
- Step 2: Temporal sampling
- Step 3: Quantization of pixel values

IMAGE CONVERSION STEPS

Step 1: Spatial sampling

- is the conversion of the continuous signal to its discrete representation called pixels
- The image is mapped onto a grid of rows and columns.
- Each small square in the grid represents a pixel.
- The number of pixels determines the spatial resolution of the image

Example

If an image is sampled at 64×64 pixels, it is divided into 4096 small picture elements. If it is sampled at 1024×1024 pixels, how many small picture elements?

IMAGE CONVERSION STEPS Cont'd

Step 2: Temporal sampling

- refers to capturing images at specific time intervals.
- It is mainly used when dealing with video or moving images.
- Instead of recording continuously, the system captures frames at discrete times

Example

A video recorded at 30 frames per second (fps) means 30 images are sampled every second.

IMAGE CONVERSION STEPS Cont'd

Step 3: Quantization of pixel values

- This is the process of assigning a finite number of intensity values to each pixel after sampling.
- The continuous brightness levels are converted into discrete numbers.
- The number of levels depends on the bit depth
- Conversion is carried out using an analog-to-digital converter, which is typically embedded directly in the sensor electronics

Example

- **8-bit image** → 256 gray levels ($0-255 = 2^8$)
- **12-bit image** → 4096 intensity levels
- **16-bit image** → ?

IMAGE CONVERSION STEPS Cont'd

Analog Image



Digitization Process

1. Spatial Sampling

Divide into Pixels



Dividing the image into a grid of pixels.

Image Grid
64 × 64 Pixels

2. Temporal Sampling

Sample in Time



— 30 FPS (Frames Per Second) —

Capturing frames at discrete time intervals.

3. Quantization of Pixel Values

Assign Intensity Levels



Mapping to discrete intensity values.

Digital Image



Digital Image

8-bit, 256 Levels

IMAGE CONVERSION



148	123	52	107	123	162	172	123	64	89	...
147	130	92	95	98	130	171	155	169	163	...
141	118	121	148	117	107	144	137	136	134	...
82	106	93	172	149	131	138	114	113	129	...
57	101	72	54	109	111	104	135	106	125	...
138	135	114	82	121	110	34	76	101	111	...
138	102	128	159	168	147	116	129	124	117	...
113	89	89	109	106	126	114	150	164	145	...
120	121	123	87	85	70	119	64	79	127	...
145	141	143	134	111	124	117	113	64	112	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

$I(u, v)$



Image details

Transformation of a continuous grayscale image $F(x, y)$ to a discrete digital image $I(u, v)$

SPECIAL IMAGES

INDEXED OR PALETTE IMAGES

- are a very special class of color image.
- The difference between an indexed image and a true color image is the number of different colors (fewer for an indexed image) that can be used in a particular image.
- The pixel values are only indices (with a maximum of 8 bits)

SPECIAL IMAGES

SPECIAL IMAGES

- required if the standard kinds of image are insufficient for representing the image values.

Two common examples of special images

- **Negative values**
(arise during image processing steps)
- **Floating-point values.**
(found in medical, biological, or astronomical applications)

CHARACTERISTICS OF SPECIAL IMAGE

Special Images:

<i>Chan.</i>	<i>Bits/Pix.</i>	<i>Range</i>	<i>Use</i>
1	16	$[-32768, 32767]$	Integer values pos./neg., increased range
1	32	$\pm 3.4 \cdot 10^{38}$	Floating-point values: medicine, astronomy
1	64	$\pm 1.8 \cdot 10^{308}$	Floating-point values: internal processing

IMAGE FILE FORMATS

- Image data must first be loaded into memory from a file.
- Files provide the essential mechanism for storing, archiving, and exchanging image data.
- The choice of the correct file format is an important decision.
- In the early days (i.e., 1985), most software developers created a new custom file format for almost every new application.
- Today, there exists a wide range of standardized file formats, and developers can almost always find at least one existing format suitable for their application.
- Using standardized file formats vastly increases
 1. The ease with which images can be exchanged and
 2. The likelihood that the images will be readable by other software in the long term.

IMAGE FILE FORMATS

FILE FORMA T	Design Purpos e	Support for Image	Strength	Weaknes s	Compress ion technique s	Applicatio n
Tagged Image File Format (TIFF)	to meet the professi onal needs of diverse fields.	grayscale, indexed, and true color images, special image types with large-depth integer and floating-point elements	enables new image types and informatio n blocks to be created by defining new “tags”	proprieta ry tags are not always supporte d flagged “unsuppo rted tag”		Archiving documents , scientific applicatio ns, digital photograp hy, and digital video production .

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakness	Compression techniques	Application
Graphics Interchange Format (GIF)	efficiently encode “flat” or “iconic” images consisting of large areas of the same color.	indexed color, grayscale images with a maximum depth of 8 bits	ability to include multiple images in a single file to create simple animations.	it does not support true color images	LZW compression, interlaced image loading	representing images on the Web

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakness	Compression techniques	Application
Portable Network Graphics (PNG)	as a universal image format to replace GIF especially for use on the Internet	true color images, grayscale images, indexed color images	it allows images of up to $2^{30} \times 2^{30}$ pixels.	supports a single image per file	uncompressed, lossless	format of choice for representing uncompressed, lossless, true color images for use on the Web

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakne ss	Compressi on techniques	Application
JPEG	Achieve an average data reductio n of a factor of 1:16	grayscale and color images	compres sion factor of around 1:25 for 24-bit color images with acceptab le quality	limitatio n to 8- bit images, its poor perform ance on non- photogr aphic images such as line art	combines a number of different compressio n methods	grayscale and color images not a file format, it is “only” a method of compressing image data

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakne ss	Compressi on techniques	Application
JPEG File Interchang e Format (JFIF)	specifies a file format based on the JPEG standard	makes a specific choice of the images in JPEG	Specifies the use of different compres sion ratios for color and luminanc e, which is a practical			grayscale and color images

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakne ss	Compressi on techniques	Application
Exchange able Image File Format (EXIF)	storing image data originati ng on digital cameras	metadata such as the type of camera, date and time, aperture and exposure time, as well as	uses TIFF to store the metadat a and JPEG to encode a thumbna il preview image			Used by all manufacture rs as the standard format for storing digital images on memory cards.

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakne ss	Compressi on techniques	Application
Windows Bitmap (BMP)		grayscale, indexed, and true color images, binary images (not efficiently)	offers storage for a similar range of image types as TIFF	it is a much less flexible format	simple lossless, run- length- based compressio n	widely used under Windows

IMAGE FILE FORMATS

FILE FORMAT	Design Purpose	Support for Image	Strength	Weakne ss	Compressi on techniques	Application
Portable Bitmap Format (PBM)	exceptio nal in that they can be optionall y saved in a human- readable text format	PBM (portable bitmap) for binary bitmaps, PGM (portable graymap) for grayscale images, and PNM (portable	much more machine- optimize d “raw” output mode in which pixel values are stored as			widely used under Unix

PRACTICAL APPLICATION OF FILE FORMAT

FILE FORMAT	PRACTICAL APPLICATION
TIFF	universal format supporting a wide variety of uncompressed images
JPEG/JFIF	digital color photos when storage size is a concern
PNG or GIF	when an image is destined for use on the Web

BIG AND LITTLE ENDIANS

- In the standard model of a computer, a file consists of a simple sequence of 8-bit bytes, and a byte is the smallest entry that can be read or written to a file.
- In contrast, the image elements as they are stored in memory are usually larger than a byte; for example, a 32-bit int value (= 4 bytes) is used for an RGB color pixel.
- Storing the four individual bytes that make up the image data can be done in different ways.

BIG AND LITTLE ENDIANS

- In order to correctly recreate the original color pixel, we must naturally know the order in which bytes in the file are arranged.

Consider, for example, a 32-bit int number z with the binary and hexadecimal values

$$z = \underbrace{00010011}_{13_{\text{H}} \text{ MSB}} \cdot \underbrace{00100101}_{86_{\text{H}} \text{ LSB}} \cdot 01110100 \cdot \underbrace{10001100}_{86_{\text{H}} \text{ LSB}}_{\text{B}} \equiv 13257486_{\text{H}},$$

BIG AND LITTLE ENDIANS

ORDERING	BYTE SEQUENCE	1	2	3	4
Big endian	MSB → LSB	13 _H	25 _H	74 _H	86 _H
<i>Little endian</i>	LSB → MSB	86 _H	74 _H	25 _H	13 _H

ASSIGNMENT

1. Determine the actual physical measurement in millimeters of an image with 1400 rectangular pixels and a resolution of 72 dpi.
2. A camera with a focal length of $f = 50$ mm is used to take a photo of a vertical column that is 12 m high and is 95 m away from the camera. Determine its height in the image in mm (a) and the number of pixels (b) assuming the camera has a resolution of 4000 dpi.
3. Determine the number of bytes necessary to store an uncompressed RGB color image of size 640×480 pixels using 8, 10, 12, and 14 bits per color channel